

HENRY AGYARE ASANTE

Dallas, Texas, USA

(214)685-8733 || com1.asantehenry@gmail.com || <https://www.linkedin.com/in/henry-asante/> || <https://github.com/henryagyare>

EDUCATION

Grambling State University (GSU)

Bachelor of Science (BSc.) Computer Science & Electrical Engineering

GPA: 4.0 / 4.0

Grambling, Louisiana

August 2024 – May 2028

Coursework: Computer Architecture, Statistics, Calculus, Data Structures and Algorithms, Data Analytics, Networking

SKILLS

- **Languages / Frameworks:** Python, Java, TypeScript, React, C/C++, Go, API, C#, Node, Kotlin, .NET, Rust, Ruby
- **Data / AI / ML:** R, SQL, TensorFlow, Pandas, Spark, GenAI, LLMs, NumPy, Matlab, Scikit-learn, PyTorch, ETL
- **DevOps / Platform:** Linux, Git, Shell, Docker, CI/CD, Kubernetes, AWS (EC2, S3, Lambda), Azure, Virtualization

WORK EXPERIENCE

Dell Technologies

Software Engineering Intern

Round Rock, Texas

May 2025 – August 2025

- Boosted hardware defect regression testing speed by 400% through an AI-driven test case recommendation system.
- Reduced defect-to-testcase mapping errors by 10% using NLP to analyze 1,200+ Jira tickets and 5,000+ testcases.
- Achieved ~96% recommendations accuracy using precision/recall metrics, ensuring high-confidence test coverage.
- Containerized a React + Flask microservice with Docker, enabling seamless deployment in Dell's CI/CD pipeline.

TL James Engineering Lab

Research / Lab Assistant

Grambling, Louisiana

September 2024 – May 2025

- Built an Arduino-based sensor payload capturing temperatures & pressures at >100,000 ft high w/ 95% accuracy.
- Optimized low-level C firmware for I²C/SPI peripherals, reducing sampling latency by 20% and ensuring reliability.
- Resolved 30+ software/hardware issues by debugging with Python, Logic Analyzers, & SPI/I²C protocol analyzers.
- Collaborated in a 10-member team through the design lifecycle: schematic drafts, PCB, calibration, and deployment.

Agisam Academy

Junior IT Specialist

Remote

January 2023 – July 2024

- Built custom automation suites using pandas, openpyxl, and SQL, streamlining student records and analytics.
- Reduced data processing time by ~120% and deployed 5 tools still being used daily across multiple departments.
- Improved org infra uptime by 60% by configuring and maintaining routers, switches, and Windows workstations.
- Provided Tier 1–2 support for Wi-Fi & OS reimaging using standard diagnostic tools & command-line utilities.

COMPETITIONS / HACKATHONS / CONTESTS

Votran — Real Time Language Learning App | (React.js, HTML, CSS, JavaScript, OpenAI Whisper, GPT API) 2025

- Led frontend development for a team of 5 building VoTran, a real-time language learning web app using React.js.
- Implemented multilingual translation and audio playback pipelines, enabling 95%+ transcription accuracy.
- Delivered full MVP in 5 hours, securing strong viability scores from judges & outperforming 10+ competing teams.

Quizler — HS/College STEM Quiz Application | (Java, SpringBoot, FastAPI, PostgreSQL, Agile, Redis) 2024

- Built a Java quiz app w/ 15.6k+ Q&A, achieving 99.9% uptime by deploying Spring Boot APIs on PostgreSQL.
- Reduced API latency by ~40% via SQL query optimization, Redis caching, & multithreading for concurrent sessions.
- Ensured 95%+ code coverage by writing JUnit/Mockito tests & collaborating in Agile sprints.

Grenes — Sustainability Focused Social App | (Kotlin, KMM, Firebase)

2024

- Collaborated to build Grenes, a cross-platform sustainability app encouraging environmentally conscious actions.
- Developed end-to-end authentication and user flows using Firebase Auth and Firestore, with <200ms login time.
- Reduced duplicated platform specific code by ~55% using KMM, accelerating feature delivery across iOS/Android.

PROJECTS

Spinal Lower Back Pain Classifier | (TensorFlow/Keras, MobileNet-V2, PyDICOM, Groq Llama 3.3) [view project](#)

- Deployed as a full-stack Streamlit app with and LLM-powered diagnostic analysis using Groq's Llama 3.3 API.
- Built and iteratively improved a CNN-based MRI classifier using MobileNet-V2, achieving ~72% training accuracy.
- Visualized accuracy & loss curves to diagnose overfitting & guide future model enhancements (fine-tuning planned).

BrillaAI — STEM Question Synthesis Platform | (Python, LoRA, LLaMA-3 8B, Unsloth, SQL, Pandas) [view project](#)

- Fine-tuned LLaMA-3 8B with LoRA and 4-bit quantization via Unsloth to generate 15,600+ STEM QA pairs.
- Increased token generation reliability for long-context outputs by 35%, & cut Tesla T4 GPU memory usage 25%.
- Built scalable pandas/SQL pipelines for ETL, ensuring reproducibility and supporting evaluation/deployment.

Orbit-sh: A Modular Unix Shell with Systems Integration | (C, Unix, Linux Systems Programming) [view project](#)

- Built a Unix shell in C supporting command parsing, foreground/background execution, & built-in commands.
- Implemented pipeline support (|) using pipe() and dup2() to enable communication between forked child processes.
- Prevented zombie processes using sigaction(SIGCHLD) w/ non-blocking waitpid(), ensuring child process reaping.